

Financial Machine Learning — Midterm Practice Set

QRM3007 • Units 1–7 • Prof. Keehong Rhee

Practice set • 48 basic questions • about 50 minutes

Instructions

- Choose exactly one answer (A–D) for each question. There is no penalty for wrong answers.
- Each question checks one basic idea. If you miss one, reread the explanation in the model answers.
- Write your name and student ID on the answer sheet.

Name: _____ Student ID: _____

Part 1 • Foundations and Financial Data (Units 1–2)

1. A bank has 50,000 past loans, and each record says whether the borrower repaid. It wants to judge new applicants. This is:

- A) No learning is needed
- B) Supervised learning
- C) Reinforcement learning
- D) Unsupervised learning

2. A marketing team has the spending records of 1 million customers but no labels. It wants to find natural customer groups. This is:

- A) Unsupervised learning
- B) Supervised learning
- C) Reinforcement learning
- D) Regression

3. A trading bot must sell a huge position. It tries many ways of slicing the order and is rewarded when trading costs are low. This is:

- A) Supervised classification
- B) Clustering
- C) Unsupervised learning
- D) Reinforcement learning

4. Team A predicts next month's return (e.g. +1.8%). Team B predicts whether a loan will default (yes/no). Their tasks are:

- A) A: regression, B: classification
- B) A: classification, B: regression
- C) Both: regression
- D) Both: clustering

5. Intern Mina's model scores 99% on the training data but only 51% on new data. Her model is:

- A) Overfitting — it memorised the training data
- B) Perfect — 99% is excellent
- C) Underfitting — it is too simple
- D) Broken — the new data must be wrong

6. Before testing, Mina must hold out data. Her bank's 10,000 loan applications come from different customers, so the order of the rows carries no information. What is the standard way to make the test set?

- A) Use all 10,000 rows for both training and testing
- B) Shuffle the rows at random, then hold out about 20% as the test set

- C) Test only on the rows the model already predicts well
 - D) Skip the test set and report the training accuracy
7. Mina's next data set is a daily time series from 2015 to 2024 — the special case. Which split gives an honest test?
- A) Train on 2023–2024, test on 2015–2022
 - B) Train and test on 2015–2024
 - C) Train on 2015–2022, test on 2023–2024
 - D) Shuffle all days, then split 80/20
8. During an earnings week a stock trades four times its usual dollar volume. Compared with a normal week, dollar bars will be:
- A) Exactly one per day
 - B) Fewer
 - C) About four times as many
 - D) The same number
9. A fund study uses a database that lists only funds still operating today. The past performance will look:
- A) Too bad, because good funds are missing
 - B) Too good, because failed funds are missing (survivorship bias)
 - C) Random
 - D) Exactly right
10. A strategy uses a company's annual earnings, published on 15 March, to trade on 1 January of the same year. This is:
- A) Regularization
 - B) Survivorship bias
 - C) Overfitting
 - D) Look-ahead bias
11. A bank has no fraud labels at all. It flags transactions that look very different from normal ones. This is:
- A) Supervised regression
 - B) Unsupervised anomaly detection
 - C) Triple-barrier labeling
 - D) Reinforcement learning

Part 2 • Labels and Linear Models (Units 3–4)

Case — one trade. You buy a stock at 100. Profit-take barrier = 104, stop-loss barrier = 96, time limit = 6 days.

12. The price path is 101, 102, 103, 104.5. What is the triple-barrier label?
- A) No label yet
 - B) +1 (the profit-take was touched first)
 - C) -1
 - D) 0
13. Another path is 99, 97, 95.5. What is the label?
- A) -1 (the stop-loss was touched first)
 - B) +1
 - C) No label yet
 - D) 0
14. A third path stays between 98 and 103 for all six days. What is the label?
- A) No label yet
 - B) -1

- C) +1
D) 0 (the time limit was reached first)
15. To fit a straight line through data points, ordinary least squares (OLS) chooses the coefficients that make what as small as possible?
A) The number of points
B) The slope
C) The sum of squared errors
D) The largest coefficient
16. An analyst tests 400 candidate factors and suspects most of them are useless. She wants the model to keep only a short list. Which method fits?
A) Adding more factors
B) Plain OLS with all 400 factors
C) Ridge (L2), which keeps every factor
D) Lasso (L1), which sets weak coefficients exactly to zero
17. 6-month and 12-month momentum are almost the same signal. In OLS their coefficients jump to +48 and -47. The analyst wants to keep both but calm them down. Which method fits?
A) Ridge (L2), which shrinks both without deleting either
B) Plain OLS
C) No method can help
D) Lasso (L1), which deletes one of them
18. A portfolio has groups of correlated factors, and the analyst wants to shrink them and also delete pure noise. The Elastic Net penalty $\lambda_1 \sum_j |\beta_j| + \lambda_2 \sum_j \beta_j^2$ fits because it:
A) Uses only the L2 penalty
B) Uses only the L1 penalty
C) Uses no penalty
D) Combines the L1 (Lasso) and L2 (Ridge) penalties
19. Match each method to its penalty: Lasso, Ridge, Elastic Net.
A) L1 + L2, L2, L1
B) L2, L1, L1 + L2
C) L1, L2, L1 + L2
D) L1, L2, no penalty

Part 3 • Precision, Recall and F1 — Three Stories

Story — airport screening. 1,000,000 passengers pass through an airport; 10 of them are terrorists. The security chief asks the data team for a model, and the team tries three versions.

20. **Model A** labels every passenger 'civilian'. Its accuracy is 99.999%, and the team is proud. What is wrong?
A) It flags too many civilians
B) Nothing — 99.999% accuracy is excellent
C) Its recall is 0% — it catches none of the 10 terrorists
D) Its precision is too high
21. The chief says: 'Catch every terrorist!' **Model B** labels every passenger 'terrorist'. What is its recall?
A) 50%
B) 100%
C) 0%
D) 0.001%
22. Model B also flags 999,990 innocent civilians. Which metric exposes this problem?

- A) No metric can show it
- B) Precision — only 10 of 1,000,000 flags are correct (0.001%)
- C) The number of passengers
- D) Recall — it is 100%

23. The team needs one score that is high only when both precision and recall are high. Which metric is that?

- A) The number of flags
- B) Accuracy

C) The F1 score, $F_1 = \frac{2PR}{P+R}$

- D) Recall alone

24. **Model C** flags 20 passengers and catches 8 of the 10 terrorists. So precision = $8/20 = 0.4$ and recall = $8/10 = 0.8$. What is its F1?

- A) 0.60
- B) 0.40
- C) About 0.53
- D) 0.80

25. Ranked by F1, which model should the chief choose?

- A) Model C — its F1 (≈ 0.53) is far above A (0) and B (≈ 0)
- B) They are all equally good
- C) Model A — it has the highest accuracy
- D) Model B — it has the highest recall

Story — COVID screening. 1,000 people are tested at a clinic; 100 are actually infected. Anyone who tests positive is sent for a confirming PCR test, so a false alarm costs little, but an infected person sent home can infect others.

	Test positive	Test negative	Total
Actually infected	90 (TP)	10 (FN)	100
Not infected	45 (FP)	855 (TN)	900

26. What is the recall (sensitivity) of this test?

- A) 0.90
- B) 0.945
- C) 0.10
- D) 0.67

27. What is the precision of this test?

- A) About 0.33
- B) 0.945
- C) 0.90
- D) About 0.67

28. The 10 infected people who tested negative were sent home. In the table they are:

- A) True positives (TP)
- B) False positives (FP)
- C) False negatives (FN)
- D) True negatives (TN)

29. Given the story (false alarms are cheap, missed cases are dangerous), the clinic should care most about:

- A) High precision — never raise a false alarm
- B) High recall — miss as few infected people as possible

- C) High accuracy only
- D) Low recall

Story — card fraud. Among 10,000 card transactions, 20 are fraud. The bank's model blocks 50 transactions, and 15 of them are really fraud.

30. The risk team asks: 'Of all the frauds, how many did we stop?' What is that number?

- A) Recall = $15 / 20 = 75\%$
- B) Precision = $15 / 50 = 30\%$
- C) $15 / 10,000 = 0.15\%$
- D) Accuracy = 99.6%

31. The call centre complains: 'Most blocked cards belonged to honest customers!' Which metric measures this complaint?

- A) The number of transactions
- B) Recall = $15 / 20 = 75\%$
- C) Accuracy
- D) Precision = $15 / 50 = 30\%$

Part 4 • Classifiers and Anomaly Detection (Units 5–6)

32. A loan officer wants the model to say 'this applicant has a 23% chance of default'. Which model gives a probability like this directly?

- A) A dollar bar
- B) K-means clustering
- C) Logistic regression
- D) Plain OLS on a 0/1 label

33. Most transactions sit in a dense crowd; one transaction sits far away in empty space. LOF (Local Outlier Factor) will:

- A) Mark the crowd as anomalies
- B) Ignore it, because it has no label
- C) Delete the whole crowd
- D) Flag it, because its local density is much lower than its neighbours'

34. Repaid customers have credit scores 2, 3, 4 and defaulters have 8, 9. Where does an SVM put the boundary?

- A) At 8
- B) At 6, the middle of the widest gap
- C) At 5
- D) At 4

35. In that example, which customers decide where the boundary goes?

- A) All five customers equally
- B) Only the closest ones, at 4 and 8 (the support vectors)
- C) Only the one at 2
- D) Only the one at 9

36. A new bank has years of normal transactions but no examples of fraud. Which model can still learn a 'fence' around normal behaviour?

- A) One-Class SVM
- B) Supervised classification with fraud labels
- C) Ordinary least squares
- D) Logistic regression

Part 5 • Trees and Ensembles (Unit 7)

Story — a credit-scoring tree. A bank builds a decision tree from 2,000 past loans to approve or reject new applicants.

37. What does a decision tree do?

- A) Averages the answers of the nearest applicants
- B) Groups applicants without using any labels
- C) Asks a series of yes/no questions (e.g. 'income > \$50k?') and gives the decision at the leaf it reaches
- D) Draws one straight line through all applicants

38. The analyst lets the tree keep splitting until every leaf holds a single applicant. Training accuracy is 100%, but accuracy on new applicants falls to 60%. What happened?

- A) Nothing — 100% training accuracy is the goal
- B) The new applicants' data are wrong
- C) Underfitting — the tree is too small
- D) Overfitting — the tree memorised the training applicants

39. To stop this, the analyst sets `max_depth = 4`. This means:

- A) Only 4 features may be used
- B) Only 4% of the data is used
- C) Exactly 4 trees are trained
- D) At most 4 questions from the root down to any leaf

40. She also sets `min_samples_split = 50`. This means:

- A) The tree is split into 50 parts
- B) A node is split only if it holds at least 50 applicants
- C) Every split uses 50 features
- D) The data are split into 50 folds

41. Next she sets `min_samples_leaf = 30`. This means:

- A) Every leaf must contain at least 30 applicants
- B) Each leaf uses 30 features
- C) The tree must have exactly 30 leaves
- D) The tree is trained 30 times

42. Finally she sets `max_leaf_nodes = 8`. This means:

- A) Each leaf holds 8 applicants
- B) The tree may end in at most 8 leaves (8 final decisions)
- C) The tree must use 8 features
- D) 8 trees vote together

Story — a random forest. Instead of one tree, the bank trains a random forest of 500 trees. Each tree learns from a random sample of the loans.

43. For a new applicant, 320 trees say 'default' and 180 trees say 'repay'. What does the forest predict?

- A) The answer of the first tree only
- B) No decision — the trees disagree
- C) Default — the majority vote (320 of 500)
- D) Repay — the minority wins

44. The bank also uses a forest to predict the loss amount (a number) if a loan defaults. How does the forest combine its 500 trees?

- A) It averages the 500 trees' predictions
- B) It multiplies the predictions
- C) It uses only the best tree
- D) It takes the largest prediction

45. A random forest belongs to a family of ensemble methods called bagging (bootstrap aggregating). What does bagging do?

- A) Trains one very deep tree on all the loans
- B) Trains trees one after another, each fixing the previous tree's mistakes
- C) Removes the labels and groups the loans
- D) Trains many trees side by side, each on a random bootstrap sample of the loans, then combines them by vote or average

46. Another team uses boosting (for example AdaBoost or gradient boosting). How does boosting build its trees?

- A) One after another: each new small tree concentrates on the loans the trees so far got wrong, and the trees are added up
- B) All at once, each on a random bootstrap sample, then a majority vote
- C) It keeps only the single best tree
- D) It trains one tree and copies it 500 times

47. Which statement about bagging and boosting is correct?

- A) Bagging and boosting are unsupervised methods
- B) Boosting trains its trees in parallel; bagging trains them in sequence
- C) Both are ensemble methods: bagging trains trees in parallel and mainly reduces variance; boosting trains them in sequence and mainly reduces bias
- D) Only boosting is an ensemble method

48. Account balances are 10, 11, 12, 13 and 45 (thousand dollars). Isolation Forest cuts the range at random. Which balance is usually isolated with the fewest cuts?

- A) 10
- B) 12
- C) All need the same number of cuts
- D) 45 — it sits alone in empty space